COMPUTER ENGINEERING

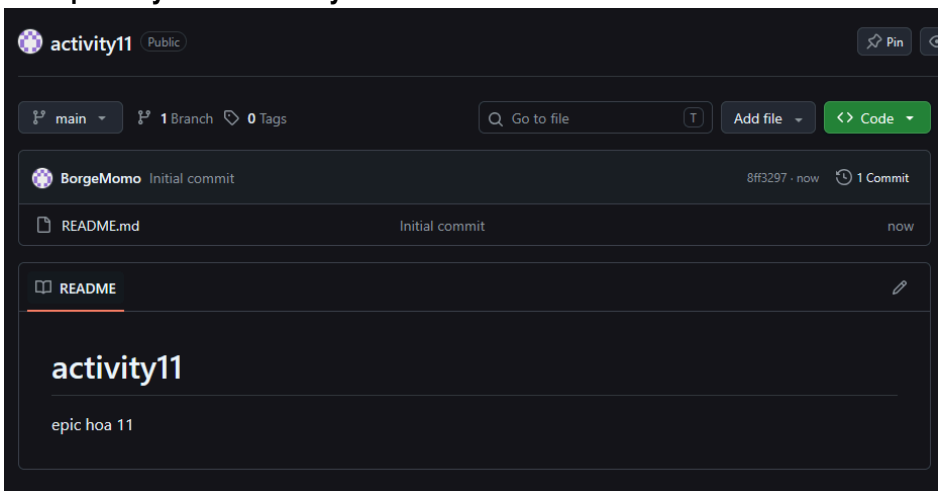
QUEZON CITY

| | | | |
|---|---|---|---|
| **COURSE CODE/TITLE:** | CPE212 - Automating Server Management | **SECTION:** | CPE31S2 |
| **DATE PERFORMED:** | 10/15/2026 | **DATE SUBMITTED:** | 10/15/2026 |
| **NAME:** | Momo, Borge Rudolf A. | **INSTRUCTOR:** | Engr. Quejado |

## ACTIVITY NO. 11.1 Containerization

## 1. PROCEDURE OUTPUT

**1. Create a new repository for this activity.**




**2. Install Docker and enable the docker socket.**

Installing Docker for UBUNTU

```
tasks:
  - name: Install Docker on Ubuntu
    ansible.builtin.apt:
      name: docker.io
      state: present
      update_cache: yes
    when: ansible_distribution == "Ubuntu"
```

```
TASK [Install Docker on Ubuntu] **************
skipping: [192.168.56.116]
changed: [192.168.56.114]
changed: [192.168.56.115]
```

Installing Docker for CentOS

```
- name: Install Docker on CentOS
  ansible.builtin.dnf:
    name: docker
    state: present
  when: ansible_distribution == "CentOS"
```

```
TASK [Install Docker on CentOS] ******
skipping: [192.168.56.115]
skipping: [192.168.56.114]
changed: [192.168.56.116]
```

Verification that Docker Socket is enabled

```
momo@workstation:~/activity11$ ansible all -i inventory.ini -b -m command -a
 "systemctl is-enabled docker.socket" --ask-become-pass
BECOME password:
192.168.56.114 | CHANGED | rc=0 >>
enabled
192.168.56.116 | CHANGED | rc=0 >>
enabled
192.168.56.115 | CHANGED | rc=0 >>
enabled
momo@workstation:~/activity11$
```

**3. Add a Docker group to your current user.**

Create a Docker Group

```
tasks:
  - name: Create the cool Docker group
    ansible.builtin.group:
      name: docker
      state: present
```

```
TASK [Create the cool Docker group] *******
ok: [192.168.56.114]
ok: [192.168.56.115]
ok: [192.168.56.116]
```

Add User to Docker Group

```
- name: Add momo to the cool Docker group
  ansible.builtin.user:
    name: momo
    groups: docker
    append: yes
```

```
TASK [Add momo to the cool Docker group] ***
changed: [192.168.56.114]
changed: [192.168.56.115]
changed: [192.168.56.116]
```

**4. Create a Dockerfile to install web and DB servers.**

Dockerfile

```
  GNU nano 7.2                                                  Dockerfile
FROM ubuntu:22.04

ENV DEBIAN_FRONTEND=noninteractive

RUN apt-get update && apt-get install -y \
    apache2 \
    mariadb-server \
    && apt-get clean \
    && rm -rf /var/lib/apt/lists/*

EXPOSE 80 3306

CMD ["bash", "-c", "service mariadb start && apachectl -D FOREGROUND"]
```

Index.html

```
  GNU nano 7.2                                                  index.html *
<!DOCTYPE html>
<html>
<head>
    <title>Activity 11 - Containerization</title>
</head>
<body>
    <h1>Welcome to Activity 11!</h1>
    <p>This webpage is running inside a Docker container.</p>
    <p>Apache2 is our web server, and MariaDB is our database server.</p>
</body>
</html>
```

**5. Install and build the Dockerfile using Ansible.**

Create Project Directory

```
tasks:
  - name: Create project directory
    ansible.builtin.file:
      path: /home/momo/activity11
      state: directory
      owner: momo
```

```
TASK [Create project directory] **
ok: [192.168.56.115]
ok: [192.168.56.114]
changed: [192.168.56.116]
```

Copy Dockerfile

```
- name: Copy Dockerfile
  ansible.builtin.copy:
    src: Dockerfile
    dest: /home/momo/activity11/Dockerfile
    owner: momo
```

```
TASK [Copy Dockerfile] *****
ok: [192.168.56.115]
ok: [192.168.56.114]
changed: [192.168.56.116]
```

Copy web Page

```
- name: Copy web page
  ansible.builtin.copy:
    src: index.html
    dest: /home/momo/activity11/index.html
    owner: momo
```

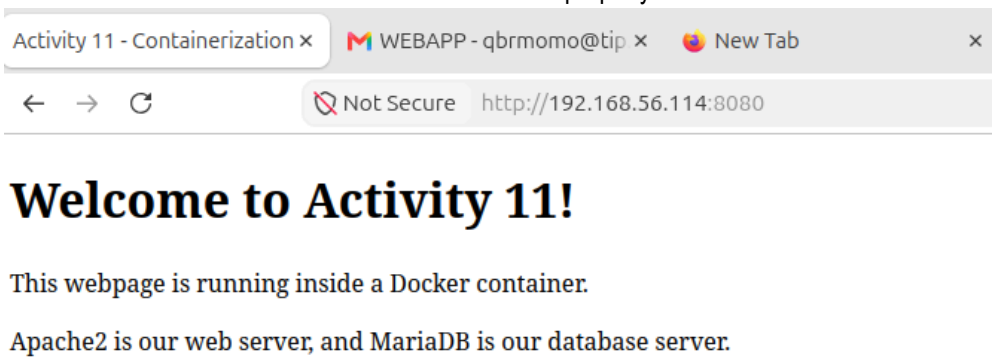
```
TASK [Copy web page] *******
ok: [192.168.56.115]
ok: [192.168.56.114]
changed: [192.168.56.116]
```

Build Docker Image

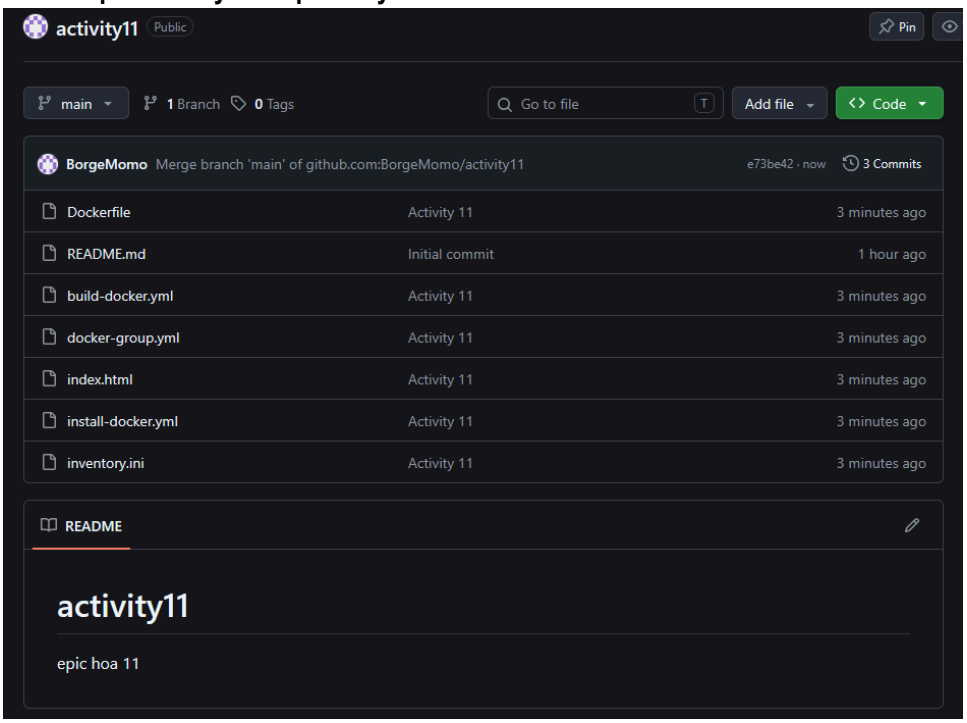
```
- name: Build Docker image
  ansible.builtin.command:
    cmd: docker build -t activity11-web-db /home/momo/activity11
```

```
TASK [Build Docker image] *******
changed: [192.168.56.114]
changed: [192.168.56.115]
changed: [192.168.56.116]
```

Verification it worked properly:




**6. Add, commit and push it to your repository**




https://github.com/BorgeMomo/activity11

COMPUTER ENGINEERING

QUEZON CITY

## 2. ANSWERS TO SUPPLEMENTAL ACTIVITY

**Reflections:**

Answer the following:

1. What are the benefits of implementing containerizations?
   - There are many benefits of containerization such as that it makes it easier to run and manage applications because everything we need is packed inside a container and it also helps applications run the same way on different computers or servers. Another benefit is that it saves time and resources because containers are lightweight and can start quickly.

## CONCLUSIONS

In conclusion, I learned how to install and manage Docker using Ansible and I also learned how to create a Dockerfile that installs Apache2 and MariaDB inside the container. What I did was build the Docker image and run a container on my server and I also tested the webpage using Firefox browser and overall, this activity was pretty easy and it helped me understand how containerization works and how it makes managing applications easier.

## 3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

**Tools Used:** **Google AI**

*Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.*

**Purpose of Use:** **Understanding of Concepts**

*Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.*

**Extent of Use:** **The tool was only used as a guide on some concepts and was not used to generate answers.**

*Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.*

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.